



Management of primary ovarian insufficiency (premature ovarian failure)

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INTRODUCTION

46,XX primary ovarian insufficiency (POI) is defined as the development of primary hypogonadism before the age of 40 years in women who have a normal karyotype. The presenting symptoms are similar to those of menopause. The age-specific incidence of spontaneous POI is approximately 1 in 250 by age 35 years and 1 in 100 by age 40 years [1]. In its fully developed form, it is associated with oligomenorrhea or amenorrhea, symptoms of estrogen deficiency, and gonadotropin levels in the menopausal range before age 40 years. The terms "premature menopause" and "premature ovarian failure" were used in the past for POI, but both are inaccurate because many patients with spontaneous POI produce estrogen intermittently and ovulate, a few experience intermittent return of regular menses, and, in 5 to 10 percent of cases, women conceive and have a normal pregnancy [2].

The management of women with POI will be reviewed here. Turner syndrome (45,X0 or other Xp chromosome abnormalities), as well as other aspects of 46,XX POI (including autoimmune POI) are reviewed separately. (See "[Clinical manifestations and diagnosis of primary ovarian insufficiency \(premature ovarian failure\)](#)" and "[Autoimmune primary ovarian insufficiency \(premature ovarian failure\)](#)" and "[Pathogenesis and causes of spontaneous primary ovarian insufficiency \(premature ovarian failure\)](#)" and "[Turner syndrome in children and adolescents: Management](#)" and "[Clinical manifestations and diagnosis of primary ovarian insufficiency \(premature ovarian failure\)](#)", section on 'Introduction'.)

INFORMING THE PATIENT OF THE DIAGNOSIS

The most important steps after making the diagnosis of spontaneous primary ovarian insufficiency (POI) ([algorithm 1](#)) are to inform the patient of the diagnosis in a sensitive and caring manner, provide accurate information, and offer referral to appropriate resources for emotional support. The most common words women use to describe their emotional state in the immediate hours after receiving the diagnosis are "devastated," "shocked," and "confused" [3].

Young women with POI are usually unprepared for the diagnosis, and the majority are unhappy with the manner in which they were informed [4]. In one study of 100 women with POI, 71 percent were dissatisfied with how they were informed of their diagnosis [3]. Specific areas of improvement suggested by women in the study included the need for clinicians to spend more time with the patient and provide more information about this condition. The diagnosis of POI brings with it the potential for development of related depression and anxiety disorders [3,5]. It is also important to provide information about the possible cause of the POI, although this is unknown in many cases. The causes of POI are reviewed separately. (See ["Pathogenesis and causes of spontaneous primary ovarian insufficiency \(premature ovarian failure\)"](#).)

It is best to schedule a return office visit to review the laboratory results when the diagnosis is suspected. Clinicians should inform patients that 50 to 75 percent of women with 46,XX spontaneous POI experience intermittent ovarian function and that 5 to 10 percent of women are able to become pregnant sometime after the diagnosis [2]. (See ["Clinical manifestations and diagnosis of primary ovarian insufficiency \(premature ovarian failure\)"](#), section on 'Clinical features' and 'Fertility' below.)

When first diagnosed with POI, some patients feel a need to act immediately to achieve a pregnancy. It is important to point out that in addition to fertility challenges, other issues need to be addressed, as POI may have adverse effects on long-term health, such as emotional health, autoimmune endocrinopathies, cardiovascular disease, and osteoporosis [6-11]. There may also be an increased cancer risk, particularly with a strong family history of cancer [12]. An increased risk of cognitive dysfunction and dementia may also be a risk, but this is not well established in women with POI not due to bilateral oophorectomy. (See ["Clinical manifestations and diagnosis of primary ovarian insufficiency \(premature ovarian failure\)"](#), section on 'Consequences of estrogen deficiency' and 'Emotional health' below and 'Autoimmune endocrinopathies' below.)

IMPORTANCE OF ESTROGEN THERAPY

Goals — Unless there is an absolute contraindication to taking estrogen therapy, all women with primary ovarian insufficiency (POI) should receive estrogen therapy to reduce the risk of osteoporosis and cardiovascular disease. In addition, estrogen therapy is important to

maintain sexual health and quality of life and to treat genitourinary syndrome of menopause (with vaginal estrogen), if necessary. Our approach is consistent with The American College of Obstetricians and Gynecologists (ACOG) Committee Opinions on POI in adolescents and young women [13] and hormone therapy in POI [14].

Although exogenous estrogen replacement is recommended, only limited data are available about the advantages or disadvantages of different hormone regimens or about estrogen's efficacy for bone health and cardiovascular disease prevention in women with POI. However, there is abundant indirect evidence from menopausal women that physiologic estrogen doses are bone protective and that the benefits of hormone therapy outweigh the risks for women in their 50s or <10 years since menopause. (See "[Clinical manifestations and diagnosis of primary ovarian insufficiency \(premature ovarian failure\)](#)", [section on 'Cardiovascular morbidity and mortality'](#) and "[Treatment of menopausal symptoms with hormone therapy](#)", [section on 'Contraindications'](#).)

Most patients are aware of the association between estrogen deficiency, the risk of osteoporosis, and the need for estrogen therapy. However, some women are initially reluctant to consider estrogen therapy based upon results from the Women's Health Initiative [WHI], a set of hormone therapy trials in postmenopausal females that reported possible risks of cardiovascular complications and breast cancer in older subjects taking hormone therapy [15]. Women should be reassured that the WHI data are not clinically relevant to young women with POI, as they are otherwise healthy and have low baseline risks of both cardiovascular disease and breast cancer. An exception would be for those with a family history of breast cancer, in whom it is suggested to stop hormone replacement at age 45 years [16]. The issue of postmenopausal hormone is discussed in detail elsewhere. (See "[Menopausal hormone therapy: Benefits and risks](#)".)

Suggested estrogen regimens — Theoretically, hormone replacement for young women with POI should mimic normal ovarian function as much as possible. [Estradiol](#) (17-beta-estradiol; E2) and micronized [progesterone](#) are bioidentical hormones, eg, they have the same molecular structure as the estradiol and progesterone produced by the ovary (see "[Preparations for menopausal hormone therapy](#)"). Optimal replacement with sex steroids depends on whether the patient presents with primary or secondary amenorrhea.

- **Primary amenorrhea** – Girls or young women with primary amenorrhea in whom secondary sex characteristics have not developed should initially be given very low doses of [estradiol](#) (at first without a progestin) in an attempt to mimic gradual pubertal maturation. (See "[Delayed puberty: Approach to evaluation and management](#)", [section on 'Estradiol'](#).)
- **Secondary amenorrhea** – Most women present with secondary amenorrhea. We initiate full replacement doses of estrogen with oral [estradiol](#) (2 mg/day) or transdermal estradiol (100 mcg patch). An estradiol vaginal ring (100 mcg daily) is another option [17-19]. This

dose is higher than what is used for postmenopausal females and is based upon the average daily production of estradiol by the premenopausal ovary.

Transdermal or vaginal delivery of [estradiol](#) have advantages over oral estradiol, including lower risks of venous thromboembolism and gallbladder disease [20,21]. However, the advantages related to vascular safety are more important in older, postmenopausal females. Therefore, for those with POI who do not like or do not tolerate transdermal or vaginal estradiol, oral estradiol is perfectly acceptable. Some women prefer to take a combined oral estrogen-progestin contraceptive as described below. (See '[COC as alternative option](#)' below.)

We do not suggest routine monitoring of serum [estradiol](#) levels [14]. We suggest starting with suggested dosing above and titrating doses to alleviate symptoms or using a lower dose that is adequate to preserve bone density in patients who experience side effects. (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on 'Dose'.)

For women with symptoms suggestive of genitourinary syndrome of menopause (vaginal dryness, urinary symptoms), we suggest adding low-dose vaginal estrogen. Many women with POI, like those with natural or surgical menopause, add low-dose vaginal estrogen, even if they are on systemic estrogen. The use of low-dose vaginal estrogen is reviewed in detail separately. (See "[Genitourinary syndrome of menopause: Treatment of vulvovaginal atrophy](#)", section on 'Vaginal estrogen therapy'.)

Choice of progestin — Most women with POI will have an intact uterus and require a progestin to prevent estrogen-induced endometrial hyperplasia and carcinoma [22]. Our first-line progestin is micronized [progesterone](#) (MP) 200 mg per day for the first 12 days of the month [19,23]. [Medroxyprogesterone acetate](#) (MPA) is used by some clinicians. For MPA, endometrial safety data come from one trial in women with POI [23] and trials in postmenopausal women. For MP, there is indirect evidence of endometrial protection from multiple trials in postmenopausal women. For women who are unable to tolerate standard progestins, another option is [estradiol](#) therapy combined with a levonorgestrel-releasing intrauterine device (IUD). For women who desire treatment for estrogen deficiency symptoms, but want to maintain the possibility of a spontaneous conception, only micronized progesterone should be used. (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on 'Progestogens'.)

Of note, there are no data comparing the effects of the transdermal estradiol-MP regimen described above with the many other hormone regimens on symptom relief, prevention of bone loss (and other diseases), quality of life, and/or sexual function. However, indirect evidence from trials in postmenopausal women suggest that there are a number of advantages to MP over MPA (no adverse effects on lipids or clotting factors), but this has not yet been studied in the POI population. (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on 'Progestogens'.)

Duration of therapy — We agree with ACOG, which recommends systemic hormone therapy until age 50 to 51 years to all women with POI [14]. An exception would be those with a family history of breast cancer, in whom it is suggested to stop hormone replacement at age 45 years [16].

The role of hormone therapy is to manage estrogen deficiency symptoms, prevent long-term health risks associated with POI (osteoporosis, coronary heart disease, stroke), improve quality of life, and maintain sexual function (some women may need vaginal estrogen in addition to systemic estrogen).

Efficacy — There is abundant indirect evidence from postmenopausal women that hormone therapy would be effective for estrogen deficiency symptoms and bone health in women with POI. While observational data suggest that early natural menopause (prior to age 44, but not POI) is associated with an increase in cardiovascular disease, data on the role of hormone therapy for cardiovascular health in women with POI are lacking. That said, we agree with ACOG that women with POI should receive estrogen therapy unless there is an absolute contraindication [14]. (See '[Importance of estrogen therapy](#)' above and '[Treatment of menopausal symptoms with hormone therapy](#)', section on '[Contraindications](#)'.)

In a systematic review and meta-analysis of 12 studies (four trials and eight cohort studies) in 806 women with POI, evidence to support bone and cardiovascular benefits of hormone therapy was limited by the use of surrogate outcomes, and heterogeneity between studies in the estrogen dose, route, and regimen) [24]. Two studies suggest that women with POI have significant vascular endothelial dysfunction, which is restored to normal by estrogen therapy [7,9].

In addition to estrogen therapy for prevention of bone loss, other important measures for bone health should be emphasized, including exercise, a healthy diet, adequate calcium and vitamin D intake, and avoiding smoking. (See '[Overview of the management of low bone mass and osteoporosis in postmenopausal women](#)'.)

COC as alternative option — A combined estrogen-progestin oral contraceptive pill (COC) containing ethinyl [estradiol](#) and a progestin is an alternative option, which has the added advantage of providing contraception should spontaneous ovarian activity resume (see '[Contraception](#)' below). Of note, however, one small trial suggested that lower-dose, physiologic hormone therapy may be more beneficial for bone mineral density than higher-dose estrogen (an oral estrogen-progestin contraceptive) [25].

Standard postmenopausal hormone therapy, which has a much lower dose of estrogen than combined estrogen-progestin contraceptives, does **not** provide effective contraception. Since spontaneous ovarian activity may resume, some form of contraception may be required if the patient is not seeking pregnancy. (See '[Contraception](#)' below.)

FERTILITY

Approximately 75 percent of women with 46,XX spontaneous primary ovarian insufficiency (POI) have potentially functional follicles remaining in the ovary. Histologic and endocrinologic evidence supports a conclusion that inappropriate follicle luteinization is the most common pathophysiologic mechanism that prevents ovulation and pregnancy in these women [26,27]. (See '[Spontaneous ovulation](#)' below.)

Although some women conceive without treatment, pregnancy rates are very low (5 to 10 percent) [2,4,28]. Options for pregnancy include in vitro fertilization (IVF) with oocyte donation, embryo donation, and adoption (see "[In vitro fertilization: Overview of clinical issues and questions](#)", section on '[When are donor oocytes used?](#)'). Some women do undergo ovulation induction with [clomiphene](#) citrate and/or exogenous gonadotropins, but we suggest against this approach as neither has been shown to be effective. Adding adjuvant medications to enhance ovulation does not improve pregnancy rates. (See '[Spontaneous ovulation](#)' below and '[Ineffective or unproven medical therapies](#)' below.)

Women will often ask about the possibility of oocyte cryopreservation at the time of impending or diagnosed POI. Oocyte cryopreservation may be indicated in women with known genetic risk for POI before ovarian function declines (fragile X messenger ribonucleoprotein 1 [*FMR1*] premutations, Turner mosaic, known autoimmune polyglandular syndrome type 2), but it is less likely to be useful at the time of the POI diagnosis based on the paucity of oocytes left in the ovary [29].

Spontaneous ovulation — In ultrasound studies of women with POI, follicular development occurs frequently, with evidence of follicle luteinization in many cases. Ovulation is infrequent, with approximately a 4 percent chance per month [26,30]. Small studies suggest frequent monitoring (every two to four weeks) for spontaneous follicle growth by ultrasound may provide an opportunity for a spontaneous conception [30]. Such an approach may also help provide closure should there be no ovarian activity documented in a three- to six-month timeframe. (See "[Clinical manifestations and diagnosis of primary ovarian insufficiency \(premature ovarian failure\)](#)", section on '[Clinical features](#)'.)

These spontaneous pregnancies progress entirely normally in most cases, and there appears to be no need for exogenous hormone supplementation during pregnancy.

It has been thought that the use of exogenous estrogen therapy might improve follicle function and therefore ovulatory rates [31]. However, in a trial of 37 women with POI receiving oral [estradiol](#) (2 mg/day) or no therapy for six weeks in a 12-week crossover design, estrogen did not appear to improve mean ovarian volume, the number or size of new follicles, or ovulatory rates [30].

Oocyte donation — Women with POI due to any cause are potential candidates for IVF with donor oocytes. In one report of 61 women with POI undergoing 90 treatment cycles, the cumulative chance of pregnancy after three cycles was approximately 90 percent [32]. Success rates for this procedure depend primarily on the age of the oocyte donor. Women with Turner syndrome require careful cardiovascular evaluation before being considered for this therapy, as deaths from aortic dissection have been reported during pregnancy. (See ["In vitro fertilization: Overview of clinical issues and questions"](#), section on 'When are donor oocytes used?' and ["Management of Turner syndrome in adults"](#), section on 'Management of fertility and pregnancy'.)

Women with POI who become pregnant by oocyte donation may have an increased risk of delivering infants who are small for gestational age and of having pregnancy-induced hypertension and postpartum hemorrhage [33-35], but these findings are controversial [36]. (See ["In vitro fertilization: Overview of clinical issues and questions"](#), section on 'When are donor oocytes used?'.)

Ineffective or unproven medical therapies

- **Clomiphene citrate and letrozole** – An anti-estrogenic agent such as [clomiphene](#) citrate and the aromatase inhibitor [letrozole](#) have not been studied in this population, but they are unlikely to be effective in women who are hypoestrogenic. (See ["Ovulation induction with clomiphene citrate"](#), section on 'Patient selection'.)
- **Gonadotropin therapy** – Exogenous gonadotropin therapy has been studied as a potential strategy to improve ovulatory rates in women with POI, but it is ineffective [2,37,38]. Exogenous gonadotropins could theoretically exacerbate unrecognized autoimmune ovarian failure [39].

Suppression of endogenous gonadotropin concentrations with pharmacologic doses of estrogen prior to gonadotropin therapy has been reported to improve ovulatory rates in some [40,41], but not all [37], studies. In the one randomized, placebo-controlled trial available, treatment with 150 mcg ethinyl [estradiol](#)/day for two weeks before and during stimulation with recombinant follicle-stimulating hormone (FSH), ovulatory rates were significantly higher in the estrogen group (32 percent, 8 of 25 women) when compared with the placebo group (0 of 25 ovulated) [41]. Ovulation only occurred in women whose serum FSH concentrations were suppressed to ≤ 15 international units/L with estrogen. It is possible the improved ovulation rates in this study were related to suppression of luteinizing hormone (LH) levels and avoidance of inappropriate follicle luteinization [31].

Suppression of endogenous gonadotropin concentrations with a gonadotropin-releasing hormone (GnRH) agonist prior to gonadotropin therapy does not appear to improve ovulatory rates [37,42,43].

- **Glucocorticoid therapy** – Glucocorticoid therapy for treatment of suspected autoimmune ovarian failure, also unproven, carries the risk of iatrogenic Cushing syndrome and osteonecrosis of the hip requiring joint replacement [44]. (See ["Autoimmune primary ovarian insufficiency \(premature ovarian failure\)"](#).)
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OTHER ISSUES

Emotional health — The diagnosis of primary ovarian insufficiency (POI) is emotionally traumatic for most women because it disrupts their life plans, hopes, and dreams with regard to future pregnancies. Women with POI may develop related depression and anxiety disorders.

As with any life-altering diagnosis, women with POI benefit from encouragement and support that helps them to regain a sense of control and confidence [5,45]. Acceptance of this diagnosis is a gradual process, and the patient needs to give herself time and permission to work through these issues. Feelings of intense grief and loss are normal. Discussions about the normal grieving process, the strains that the diagnosis might bring to a relationship, and the importance of open communication can be helpful. For women who experience significant anxiety and/or depression related to the diagnosis, we suggest seeing a therapist with expertise in POI. In some cases, ongoing individual or group counseling is helpful [45].

Some young women interpret the diagnosis of POI as an indication that they have become a menopausal woman who is "growing old overnight" [45]. Symptoms of estrogen deficiency such as hot flashes and vaginal dryness can be interpreted as signs of premature aging. However, as noted, estrogen therapy can prevent some of the adverse consequences of estrogen deficiency. (See ["Goals"](#) above.)

Sexual health

Vaginal estrogen — Most women who are taking a full replacement dose of physiologic [estradiol](#) score within the normal range on a validated measure of sexual function. Only 7 percent score below the second percentile on a composite sexual function score [46]. As a group, however, compared with controls, women with 46,XX POI score adversely with regard to sexual function [46,47]. Low-dose vaginal estrogen therapy is an option for those who are diagnosed with genitourinary syndrome of menopause. (See ["Overview of sexual dysfunction in females: Epidemiology, risk factors, and evaluation"](#) and ["Genitourinary syndrome of menopause: Treatment of vulvovaginal atrophy"](#), section on 'Vaginal estrogen therapy'.)

Cancer survivors — Women with POI due to cytotoxic drugs or radiation therapy are often candidates for [estradiol](#) therapy. However, the decision to treat with estradiol depends upon the type of cancer. As an example, systemic (but not low-dose vaginal estrogen) is

contraindicated in women with breast cancer; conversely, estrogen is prescribed for women with Hodgkin lymphoma and ovarian insufficiency to preserve bone health and prevent cardiovascular disease. Like women with other causes of POI, the current approach is to continue estradiol until approximately age 50 years, the average age at natural menopause.

Role of androgen replacement — We suggest against the routine use of androgen therapy in women with POI. Although women with POI may have some degree of androgen deficiency when compared with young women without ovarian insufficiency, [testosterone](#) therapy has not been shown to be beneficial, and it is associated with important side effects.

In several studies of women with POI, serum ovarian androgen concentrations (androstenedione and/or [testosterone](#)) were lower than age-matched women without ovarian insufficiency, but similar to those seen in older postmenopausal women [48-50]. In contrast, levels of dehydroepiandrosterone sulfate (DHEAS), an adrenal androgen, were normal (although they would be expected to be low in women with coexisting primary adrenal insufficiency).

The clinical consequences of this decrease in ovarian androgens and the possible role of androgen therapy in women with POI have not been extensively studied. However, available data suggest that [testosterone](#) therapy is not beneficial. In one trial, 128 women with 46,XX POI on estrogen-progestin therapy were randomly assigned to 12 months of transdermal testosterone (150 mcg patch) or placebo [51]. In spite of a significant increase in mean serum total testosterone concentrations in the testosterone group compared with placebo (50 ng/dL and 18 ng/dL at 12 months, respectively), there were no differences in quality of life, self-esteem, or mood symptoms between the two groups. In a second study, the addition of testosterone to three years of estradiol-progestin therapy did not provide added benefit in bone mineral density [23].

Potential side effects of androgen replacement include hirsutism and acne and, with oral preparations (eg, dehydroepiandrosterone [DHEA]), dyslipidemia. However, in women with autoimmune ovarian failure and coexisting adrenal insufficiency, adrenal androgen therapy with DHEA may be beneficial.

Contraception — Women with POI should be informed that [estradiol/progesterone](#) replacement therapy regimens do not provide effective contraception. Therefore, since spontaneous ovarian activity may resume, contraception is required for those who are not pursuing pregnancy [14]. Women who are concerned about the possibility of spontaneous ovulation and pregnancy can choose hormonal contraception or a barrier method of contraception [19,52,53]. (See '[COC as alternative option](#)' above.)

ACOG suggests combined oral estrogen-progestin contraceptives. Other contraceptive options include lower, noncontraceptive doses of [estradiol](#) (to treat estrogen deficiency

symptoms and provide protection against future disease) combined with a [levonorgestrel](#) intrauterine contraceptive device (IUD) or oral [norethindrone](#) 0.35 mg/day (a contraceptive dose of progestin).

Other contraceptive options are reviewed separately. (See "[Contraception: Counseling and selection](#)".)

Autoimmune endocrinopathies — Young women with POI are at increased risk for developing other autoimmune endocrinopathies including autoimmune adrenal insufficiency, a potentially fatal disorder [54]. If proper screening with adrenal autoantibodies is performed, approximately 3 percent of women with spontaneous POI will be found to have asymptomatic autoimmune adrenal insufficiency ([table 1](#)) [55]. (See "[Clinical manifestations and diagnosis of primary ovarian insufficiency \(premature ovarian failure\)](#)", section on 'Test for adrenal autoantibodies'.)

- **Women with positive adrenal autoantibodies** – Even if their adrenal function is found to be normal at initial evaluation, women with positive adrenal antibodies should be followed annually by these tests. The author of this topic does an annual 8 AM serum cortisol. If <15 mcg/L, a corticotropin (ACTH) stimulation test is necessary.
- **Women with negative adrenal autoantibodies** – There are no data to inform initial adrenal function assessment and follow-up for patients with negative tests for adrenal autoimmunity. If adrenal autoantibodies are not present in a woman with POI, a reasonable strategy is to test baseline adrenal function at the time of diagnosis, and if normal, repeat the test only if clinical symptoms develop that suggest adrenal insufficiency (ie, increased skin pigmentation, excessive fatigue, orthostatic hypotension, etc). (See "[Clinical manifestations and diagnosis of primary ovarian insufficiency \(premature ovarian failure\)](#)", section on 'Test for adrenal autoantibodies'.)
- **Autoimmune hypothyroidism** – Young women with spontaneous POI are at increased risk for developing autoimmune hypothyroidism and should therefore have a yearly serum thyroid-stimulating hormone (TSH) measurement [56]. (See "[Clinical manifestations and diagnosis of primary ovarian insufficiency \(premature ovarian failure\)](#)", section on 'Other autoimmune evaluation'.)

WOMEN WITH AUTOIMMUNE POI

The management of women with autoimmune primary ovarian insufficiency (POI) is the same as for any woman with POI. Management focuses on (see "[Autoimmune primary ovarian insufficiency \(premature ovarian failure\)](#)"):

- Consequences of estrogen deficiency (vasomotor symptoms, vaginal atrophy, osteoporosis, and a possible increased risk of coronary heart disease and stroke if not treated with estrogen)
 - Emotional health/psychosocial support
 - Contraception and fertility, which is dramatically reduced
 - Evaluation for other autoimmune disorders ([table 1](#))
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SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "[Society guideline links: Primary ovarian insufficiency](#)".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "[Patient education: Primary ovarian insufficiency \(The Basics\)](#)")
 - Beyond the Basics topics (see "[Patient education: Primary ovarian insufficiency \(Beyond the Basics\)](#)")
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SUMMARY AND RECOMMENDATIONS

- **Informing the patient** – The management of women with spontaneous primary ovarian insufficiency (POI) begins with informing the patient of the diagnosis in a sensitive and

caring manner, providing accurate information about the diagnosis, and referring women with the condition to appropriate resources for emotional support ([algorithm 1](#)).

Young women with this disorder are unprepared for the diagnosis, and unfortunately, most are dissatisfied with the manner in which their health care providers delivered the information. (See '[Informing the patient of the diagnosis](#)' above and '[Emotional health](#)' above.)

- **Estrogen therapy** – Unless there is an absolute contraindication to taking estrogen therapy, we suggest systemic [estradiol](#) therapy for women with POI (**Grade 2C**) to reduce the risk of osteoporosis, cardiovascular disease, and urogenital atrophy and to maintain sexual health and quality of life. A progestin needs to be added for those with an intact uterus. (See '[Importance of estrogen therapy](#)' above and '[Choice of progestin](#)' above.)
- **Duration of therapy** – We suggest that estrogen therapy be continued at least until approximately age 50 to 51 years, the average age of natural menopause (**Grade 2C**). (See '[Importance of estrogen therapy](#)' above.)
- **Estrogen-progestin contraception** – Estrogen-progestin contraception is an alternative way to provide estrogen for women in whom pregnancy prevention is a priority. (See '[Suggested estrogen regimens](#)' above.)
- **Infertility** – For the anovulatory infertility associated with POI, we suggest not using ovulation induction drugs as they have no proven benefit (**Grade 2C**).

In vitro fertilization (IVF) with donor oocytes or donor embryos, given the high success rates, may be an option. (See '[Fertility](#)' above.)

- **Monitoring for hypothyroidism** – Young women with spontaneous POI are at increased risk for developing autoimmune hypothyroidism and should therefore have a yearly serum thyroid-stimulating hormone (TSH) measured. (See '[Autoimmune endocrinopathies](#)' above.)
- **Women with autoimmune oophoritis** – Women with positive adrenal antibodies but normal adrenal function at the initial evaluation (corticotropin [ACTH] stimulation test) should be retested annually. The author of this topic performs an annual 8 AM serum cortisol. If serum cortisol is <15 mcg/L, an [ACTH stimulation test](#) should be performed. If adrenal autoantibodies are not present in a woman with POI, a reasonable strategy is to test baseline adrenal function at the time of diagnosis, and if normal, repeat the test only if clinical symptoms develop that suggest adrenal insufficiency (ie, increased skin pigmentation, excessive fatigue, orthostatic hypotension, etc). (See '[Autoimmune primary ovarian insufficiency \(premature ovarian failure\)](#)', section on '[Additional autoimmune evaluation after diagnosis](#)' and '[Autoimmune endocrinopathies](#)' above.)

- **Turner syndrome** – The management of women with Turner syndrome (45,X0) is reviewed separately. (See "[Turner syndrome in children and adolescents: Management](#)".)
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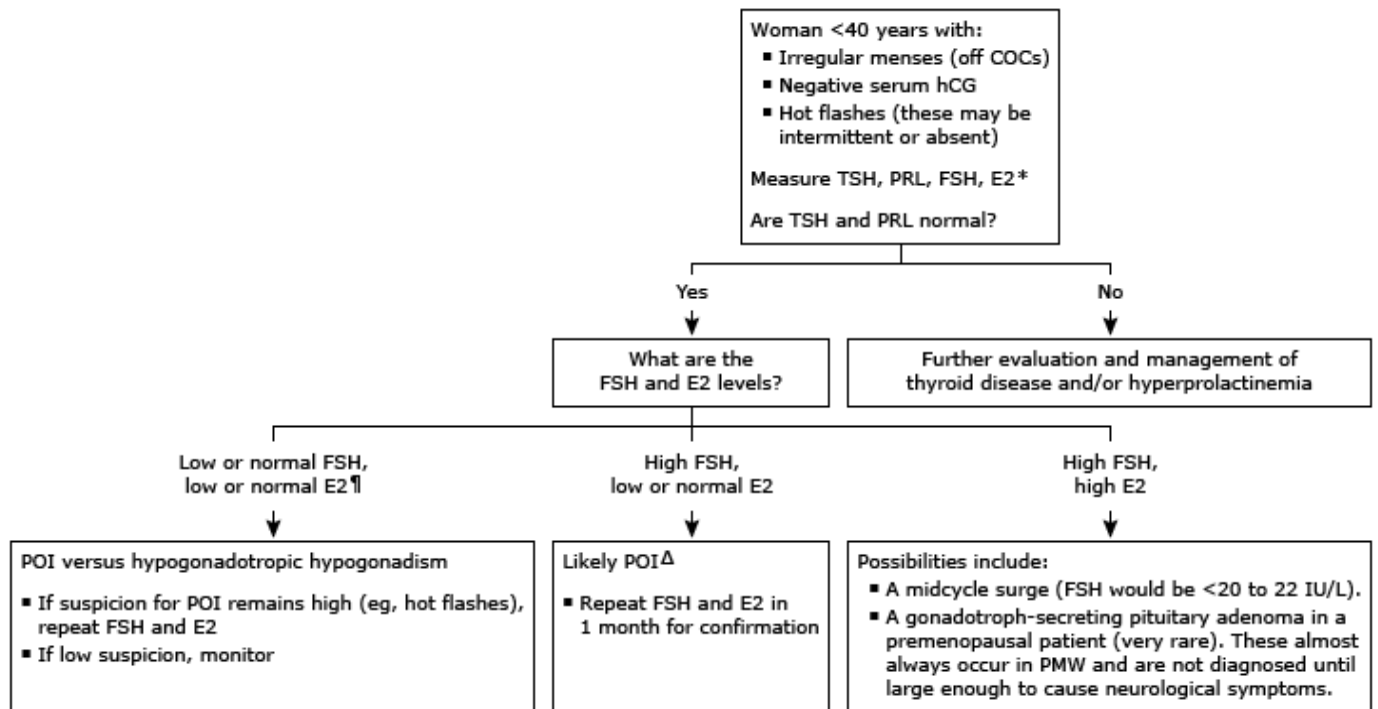
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GRAPHICS

Algorithm 1: Diagnosis of primary ovarian insufficiency



COCs: combined estrogen-progestin oral contraceptives; hCG: human chorionic gonadotropin; TSH: thyroid-stimulating hormone; PRL: prolactin; FSH: follicle-stimulating hormone; E2: estradiol; POI: primary ovarian insufficiency; PMW: postmenopausal women.

* If patient is having menses, draw labs on day 3 of a cycle (important for FSH); if not, draw labs on a random day.

¶ Low/normal FSH and E2 can be seen in women with functional hypogonadotropic hypogonadism/functional hypothalamic amenorrhea. However, hot flashes are very uncommon in this population. This same FSH/E2 pattern may be seen in women with POI who have recently ovulated. If intermittent menstrual dysfunction and/or hot flashes persist, POI is likely, and can be confirmed with repeat labs. Refer to UpToDate content on primary ovarian insufficiency and functional hypothalamic amenorrhea.

Δ Criteria for POI diagnosis: FSH >25 IU/L on 2 measurements, irregular menses for greater than 4 months, without another endocrine disorder (thyroid disease, hyperprolactinemia). Of note, some women with POI first present with infertility. Refer to related UpToDate content on primary ovarian insufficiency.

Table 1: Types of endocrine and nonendocrine autoimmune syndromes associated with adrenal insufficiency

Disorder	Prevalence (%)
Polyglandular autoimmune syndrome type 1	
Endocrine	
Hypoparathyroidism	89
Chronic mucocutaneous candidiasis	75
Adrenal insufficiency	60
Primary hypogonadism	45
Hypothyroidism	12
Type 1 diabetes mellitus	1
Hypopituitarism	<1
Diabetes insipidus	<1
Nonendocrine	
Malabsorption syndromes	25
Alopecia totalis or areata	20
Pernicious anemia	16
Chronic active hepatitis	9
Vitiligo	4
Polyglandular autoimmune syndrome type 2	
Endocrine	
Adrenal insufficiency	100
Autoimmune thyroid disease	70
Type 1 diabetes mellitus	50
Primary hypogonadism	5 to 50
Diabetes insipidus	<1
Nonendocrine	
Vitiligo	4
Alopecia, pernicious anemia, myasthenia gravis, immune thrombocytopenia purpura, Sjögren syndrome, rheumatoid arthritis	≤1

Data from:

1. Leshin M. Polyglandular autoimmune syndromes. *Am J Med Sci* 1985; 290:77.

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